



HK IMO Training 2023-2024
Problem Set 8
For 25 November 2023



Problem 29. Find all positive integers n such that there exists a positive integer k satisfying the following property: for every positive divisor d of n , the integer $d - k$ is also a (not necessarily positive) divisor of n .

Problem 30. Let $ABCD$ be a rhombus. Consider two movable points P and Q on the sides AB and AD respectively, such that PQ is tangent to the inscribed circle of $ABCD$. Prove that the area of $\triangle CPQ$ is a constant independent of the positions of P and Q .

Problem 31. Let $n \geq 4$ be a positive integer. There are n boys and n girls sitting around a circular table. Two girls will chat with one another if one of the following holds:

- there is at most one person sitting between them; or
- there are exactly two people sitting between them, at least one of whom is a boy.

Prove that there are at least n pairs of girls who will chat with one another no matter how the people are seated.

Problem 32. Find the smallest real number c such that

$$\sum_{k=1}^n \left(\frac{1}{k} \sum_{j=1}^k x_j \right)^2 \leq c \sum_{k=1}^n x_k^2$$

for all positive integers n and positive real numbers $x_1, x_2, \dots, x_n$.